Total No. of Questions : 8]

SEAT No. :

PC2632

[Total No. of Pages : 2

[6354]-781

B.E. (AIDS)

MACHINE LEARNING

(2019 Pattern) (Semester - VII) (417521)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
2) *Neat diagrams mus be drawn wherever necessary*
3) *Figures to the right indicates full Marks*
4) *Use of electronic pocket calculator is allowed*

Q1) a) What are kernel functions in SVM? Describe the Radial Basis Kernel, and Sigmoid kernel. **[6]**

b) Write formula for accuracy, precision, recall and fl-score. Calculate accuracy, precision, recall and fl-score for given example. **[12]**

		Actual Values	
		Cancer	No Cancer
Predicted Values	Cancer	45	18
	No Cancer	12	25

OR

Q2) a) Explain any 4 evaluation measures of Multiclass classification. **[4]**

b) Differentiate Balanced and Imbalanced Classification. **[6]**

c) Write a detailed note on K Nearest Neighbour algorithm with suitable example. **[8]**

Q3) a) Differentiate Agglomerative Hierarchical Clustering and Divisive Hierarchical Clustering. **[8]**

b) What is clustering? Elaborate Types of Clustering. **[9]**

OR

Q4) a) Explain DBSCAN algorithm with advantages and disadvantages. **[8]**

b) Describe centroid based clustering algorithm and explain any one type with example. **[9]**

P.T.O.

Q5) a) Explain Following with respect to Ensemble learning. **[9]**

i) Need of Ensemble Learning

ii) Advantages of Ensemble methods

iii) Ensemble learning limitations

b) Elaborate stacking approach of ensemble with example. **[9]**

OR

Q6) a) Describe ensemble learning. Explain Gradient Boosting ensemble learning techniques. **[9]**

b) Explain any three voting mechanism in ensemble learning. **[9]**

Q7) a) Differentiate supervised and unsupervised learning with example. **[8]**

b) Explain Reinforcement Learning need and its types in detail? **[9]**

OR

Q8) a) Explain following terms: **[8]**

i) Belman Equation

ii) Markov Chain

iii) Q table

iv) Q function

b) How does the Markov property relate to Reinforcement Learning? Why is it important? **[9]**

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